

ICBO 2026 - Talks by Day and Time

Author(s), title, and scheduled time for each talk

Presenting speaker shown in brown

Monday, August 3

Morning Session I | 10:30 AM - 12:00 PM

SESSION 1: Bilevel Optimization with Multiple Leaders or Followers (4 Talks)	SESSION 2: Robust and Stochastic Problems (4 Talks)	SESSION 3: Bilevel Optimization for the Electricity Sector 1 (4 Talks)
Hori, Atsushi; Okuno, Takayuki; Fukuda, Ellen H. Utilization of the Regularized Nikaido-Isoda Function for Multi-Leader-Follower Games Beyond Followers' Convexity Svensson, Anton; Salas, David Existence of solutions for deterministic bilevel games under a general Bayesian approach Yu, Xinyue; Liu, Shaonan; Yih, Yuehwern; Kong, Nan Government Subsidy Optimization for Redesigning Mutually Disjoint for-Profit Service Networks – A Single-Leader Multi-Follower Bilevel Programming Approach Flocco, Dominic; Gabriel, Steve; Schiewe, Philine A Nested Benders Decomposition Framework for Multi-Follower Bilevel Optimization	Chen, Shuyi; Zhu, Shixiang; Sioshansi, Ramteen Large-Scale Resilience Planning for Wildfire-Prone Electricity-System via Adaptive Robust Optimization Gu, Wei; Chen, Shuyi; Zhu, Shixiang; Sioshansi, Ramteen Multistage Adaptive Robust Optimization for Irreversible Long-term Resilience Planning Ebrahimi, Mohammadjavad; Shanbhag, Uday V.; Yousefian, Farzad On the Resolution of Stochastic MPECs over Networks: Distributed Implicit Zeroth-Order Gradient Tracking Methods Soto-Vergel, Angelo; Walteros, Jose; Ramirez-Rios, Diana; Borrero, Juan A Bilevel Stochastic Optimization Framework for Strategic Deployment of Survivor Geolocation Systems and Rescue Teams in Post-Disaster Operation	Chen, Yize Learning-Augmented Stackelberg Games for Scalable Demand Response in Electricity Markets Cai, Chengyi; Zhang, Peter; Zhu, Woody Two-Stage Stochastic Optimization for Reliable and Grid-Compatible Expansion of AI Data Centers Bulaich Mehamdi, Abdellah; van Ackooij, Wim; Brotcorne, Luce; Gaubert, Stéphane; Jacquet, Quentin Electricity Pricing: A Bilevel Programming Approach for Time-of-Use Pricing and Flexible Consumers Rojo, Luis; Anjos, Miguel F.; Brotcorne, Luce; Gendreau, Michel The Charging Facility Location-and-Operation Problem

Afternoon Session I | 1:30 PM - 3:00 PM

SESSION 4: Solvers and Instances (3 Talks)	SESSION 5: Nonlinear Bilevel Optimization (3 Talks)	SESSION 6: Location Problems (3 Talks)
Thürauf, Johannes; Kleinert, Thomas; Ljubić, Ivana; Ralphs, Ted; Schmidt, Martin BOBILib: Bilevel Optimization (Benchmark) Instance Library Bolusani, Suresh; Battista, Federico; Ralphs, Ted GUMBO: A Software Framework for Generalized Benders' Decomposition of Mixed-Integer Bilevel and Two-Stage Stochastic Optimization Ashraf, Mahmoud; Lamperski, Jourdain; Prokopyev, Oleg Generalizing single-level relaxations for bilevel linear programs	Song, Zitao; Padmanabhan, Swati; Zhang, Zhe Goldstein Piecewise Smooth Exploration for Linearly Constrained Bilevel Optimization Dey, Soumitra A New Class of Convex Functions and Characterizations Sharifi, Sina; Yazdandoost Hamedani, Erfan; Fazlyab, Mahyar Sequential QCQP for Bilevel Optimization with Line Search	Almeida Silva, Warley; Carvalho, Margarida; Jena, Sanjay Dominik Stackelberg Dynamic Location Planning under Cumulative Demand Ju, Jiachen; Zeng, Bo; Zeng, Shangzhi; Zhang, Jin Solving A Discrete Bilevel Healthcare Facility Location Problem via Submodular Maximization and Branch-and-Cut Choi, Yeonjun; Zeng, Bo; Lee, Kangbok Exact Algorithms for Robust Container Pre-Marshalling and Relocation under Retrieval-Priority Uncertainty

Afternoon Session II | 3:30 PM - 5:00 PM

SESSION 7: Uncertainty in Bilevel Optimization 1 (3 Talks)	SESSION 8: Bilevel and Inverse Optimization 1 (3 Talks)	SESSION 9: Equilibria and Bilevel Optimization (3 Talks)
Cotrina, John; Flores, Gonzalo; Salas, David; Svensson, Anton Stability of expectation with decision-dependent uncertainty over moving sets Vasquez, Sebastian; Bodur, Merve; Pagnoncelli, Bernardo Bridging Stochastic and Bilevel Optimization: A Unified Framework for Hierarchical Decisions under Uncertainty Xie, Yu; Ralphs, Ted K; Prokopyev, Oleg Robustness and Bounded Rationality in Mixed Integer Bilevel Linear Optimization	Zhou, Wenbin; Orfanoudaki, Agni; Zhu, Shixiang Conformalized Decision Risk Assessment Wu, Guanting; Zhang, Peter Learning from Experts under Decision Coupling: An Application to Robust Disaster Response Holani, Anurag; Gupta, Rishabh; Zhang, Qi An inverse mixed-integer optimization framework for learning interpretable models of expert decision-making	Gautam, Pankaj Forward-backward-forward dynamics for bilevel equilibrium problem Lin, Yen-Cherng Set Equilibrium Problems with Bilevel Structures Zhu, Quanyan Bilevel Cooperative Dynamics and Punctuated Equilibrium: From Tactical Consensus to Strategic Coalition Formation

Tuesday, August 4

Morning Session I | 8:30 AM - 10:00 AM

SESSION 10: Mixed-Integer Bilevel Optimization (4 Talks)	SESSION 11: Uncertainty in Bilevel Optimization 2 (4 Talks)	SESSION 12: Selected Applications of Bilevel Optimization 1 (3 Talks)
Machado Pacheco, Bruno; Carvalho, Margarida; Yu, Kim Valid Inequalities for Binary Bilevel Problems with Lower-Level Graph Reformulations Feleke, Flagot; Kassa, Semu Exact Solution Approach for Bi-level Mixed Integer Linear Fractional Programming Problems and its application in Sustainable Supply Chain for Agricultural Products Vasquez, Sebastian; van Hoeve, Willem-Jan; Lozano, Leonardo Network Flow-based Cuts for Binary Linear Bilevel Optimization Titus, William; Lunday, Brian; Robbins, Matthew; Hopkinson, Kenneth The Network Fortification Problem for Evacuation Operations	Sinha, Ankur Bilevel Optimization under Uncertainty: Applications in Network Interdiction Tian, Yiqi; Zeng, Bo A Computational Study on Distributionally Robust Optimization Bienstock, Daniel; Sisson, Blake Derisking solutions to optimization problems Beck, Yasmine; Ljubić, Ivana; Yan, Wenjin Robust Combinatorial Optimization with Decision-Dependent Budget Uncertainty	Whitehead, Nathan Network Interdiction of Disaster Response for Rapid Distribution and Maximal Covering of Demands Zhu, Xiaoyue; Adulyasak, Yossiri; Xia, Wenyi Joint Pricing and Capacity Investment Problem for Cooperative Port Authorities Sahu, Rishabh When Priority Fails: A Bilevel Framework for Resolving Passenger-Freight Conflicts in Indian Railways

Morning Session II | 10:30 AM - 12:00 PM

SESSION 13: Uncertainty in Bilevel Optimization 3 (4 Talks)	SESSION 14: Selected Applications of Bilevel Optimization 2 (4 Talks)	SESSION 15: Bilevel and Inverse Optimization 2 (4 Talks)
Hao, Hao; Brennan, Mark; Zhang, Peter Protecting Senegal's Road Network Against Flood Risk and Policy Uncertainty Sahu, Vishnu; Zhang, Qi Robust sequential bilevel programming with unknown follower's objective Perez Tezoco, Jaime; Borrero, Juan S.; Lozano, Leonardo A Bilevel Optimization Approach for a Class of Combinatorial Problems with Correlated Disruptions and Probing Liu, Suri; Zhang, Yiling; Basciftci, Beste; Wang, Wenyan Stochastic Bilevel Optimization for the Network Design of Multimodal Transit Systems	Deng, Haoyun; Wei, Peng; Xie, Weijun Bilevel Risk-Averse Routing and Resource Allocation for Urban Air Mobility Miao, Yidi; Han, Eojin; Zhang, Peter Joint Spatial Partitioning and Dispatch via Bilevel Optimization for Last-Mile Delivery Sapirstein, Abel MOMS: A Model for Optimizing Maternal Systems of Risk-Appropriate Care Song, Mingyao Bi-Level Predict-then-Optimize for Contract Interactions in Livestream Shopping	Ley, Eva; Merkert, Maximilian On Integral Inverse Optimization Problems Nadi, Sina; Lee, Taewoo; Prokopyev, Oleg A. On the inverse optimal value problem for a class of bilevel linear programs Seidel, Tobias Solving inverse robust optimization problems with bilevel-accelerated adaptive discretization methods Ota, Matheus; Vasquez, Sebastian A Geometric Perspective on Inverse Combinatorial Optimization via Adjacency

Afternoon Session I | 1:30 PM - 3:00 PM

SESSION 16: Bilevel Optimization and AI Systems (3 Talks)	SESSION 17: Learning and Bilevel Optimization 1 (3 Talks)	SESSION 18: Bilevel Optimization for the Electricity Sector 2 (3 Talks)
Dumouchelle, Justin; Julien, Esther; Kurtz, Jannis; Khalil, Elias Neur2BiLO: Neural Bilevel Optimization Zhu, Quanyan Generative LLM Signaling Games: Censorship Evasion, Semantic Blindness, and Awareness Shaping Garee, Isaac; Giovannelli, Tommaso A stochastic algorithm for bilevel optimization with a mixed-integer upper-level problem and applications to neural architecture search	Zhao, Tongke; Ghadimi, Saeed; Wolkowicz, Henry Preconditioned Bilevel Optimization Elrefaei, Jessica; Hua, Kaixun; Tran, Hoang; Borrero, Juan Graph4BiLO: Graph-Based Value Function Approximation for Bilevel Optimization Zhang, Yilang; Li, Bingcong; Giannakis, Georgios B. Learnable Loss Geometries with Mirror Descent for Scalable and Convergent Meta-Learning	Nguyen, Dai Nam; Bhuiyan, Tanveer; Zeng, Bo; Chen, Yang A Stochastic Game-Theoretic Optimization Approach for Managing Coupled Local Electricity and Thermal Markets with Electric Vehicles and Renewable Sources Andrew, Mastin A Formulation for Power Flow Interdiction with Bounded Big-M Values Kazi, Saif; Sundar, Kaarthik On the Choice of Single-Level Reformulations for Bilevel Optimization in Optimal Power Flow Networks

Wednesday, August 5

Morning Session I | 8:30 AM - 10:00 AM

SESSION 19: Selected Applications of Bilevel Optimization 3 (4 Talks)	SESSION 20: Uncertainty in Bilevel Optimization 4 (4 Talks)	SESSION 21: Interdiction Problems (4 Talks)
Arango Marin, Jaime; Bustamante Gutierrez, Lina Clemencia; Sierra Narvaez, Francisco Javier Advanced time series modeling for forecasting imbalances in maritime transport using bi-level optimization Sheldon, Andrew; Lunday, Brian; Caballero, William The Location Set Covering Disruption Problem Vo, Dung; Tran, Dat; Khanduri, Prashant Bilevel Optimization under Non-Convex Non-Smooth Settings Manimangalam, Praveen A Bilevel Optimization Framework for Self-Configuring Agentic CRM Architectures in Small-Scale Retail Enterprises	Hao, Hao; Zhang, Peter Robust Paths: Geometry and Computation Thürauf, Johannes; Lefebvre, Henri Robust Bilevel Optimization with a Wait-and-See Follower: A Column-and-Constraint Generation Approach Beck, Yasmine; Büsing, Christina; Ljubić, Ivana Exact Methods for Solving k-Delete Recoverable Robust 0-1 Problems Under Budgeted Uncertainty	Huntington, Rie; Zeng, Bo Bilevel Mixed-Integer Interdiction of the Shortest Path Problem Truong, Tran Mai Anh; Lozano, Leonardo; Sefair, Jorge A. Network Interdiction with General Integer Variables Dadpour, Amir; Fukasawa, Ricardo; Weninger, Noah Improved Algorithms for Shortest-Path Network Interdiction Zhou, Bo; Jiang, Ruiwei; Shen, Siqian Submodularity and Supermodularity in Min-Cost Flow Interdiction

Morning Session II | 10:30 AM - 11:40 AM

SESSION 22: Single-Level Reformulations (3 Talks)	SESSION 23: First-Order Methods (3 Talks)	SESSION 24: Structural and Multiobjective Perspectives in Bilevel Optimization (3 Talks)
Zemkoho, Alain True single-level reformulation for pessimistic bilevel optimization Singh, Vinay Optimality Conditions for Nondifferentiable Interval-Valued Bilevel Optimization Problems via Tangential Subdifferentials Clemente, Eneko; Prokopyev, Oleg Dimensionality Reduction in Bilevel Linear Programming	Chen, Lesi First-Order Bilevel Optimization: Computability, Lower Bounds, and Near-Optimal Algorithms Bohne, Jason; Rosenberg, David; Kazantsev, Gary; Polak, Pawel Bregman Geometry for Stochastic Online Bilevel Optimization Ji, Kaiyi Lower Complexity Bounds for Nonconvex-Strongly-Convex Bilevel Optimization with First-Order Oracles	Chen, Lisha Optimization over the Pareto set beyond convex objectives Ralphs, Ted; Fallah, Samira; Battista, Federico The Relationship Between Bilevel Optimization and Multiobjective Optimization Weninger, Noah; Fukasawa, Ricardo General framework for relaxing some Stackelberg zero-sum games